



**C20-EC-501**

**7640**

**BOARD DIPLOMA EXAMINATION, (C-20)**

**MARCH/APRIL—2025**

**DECE – FIFTH SEMESTER EXAMINATION**

**INDUSTRIAL MANAGEMENT AND SMART TECHNOLOGIES**

*Time : 3 Hours ]*

*[ Total Marks : 80*

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**PART—A**

**3×10=30**

**Instructions :** (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Define scientific management.
2. Differentiate between Authority and Responsibility.
3. Mention the merits and demerits of Trait theory of leadership.
4. Define noise reduction.
5. List types of documentation.
6. List out rules of preparation of effective document.
7. Write about the importance of start ups.
8. State the financial assistance programmes by Central Governments.
9. List out the key features of Internet of Things (IoT).
10. State the advantages and disadvantages of Internet of Things (IoT).

**PART—B**

8×5=40

- Instructions :** (1) Answer *any five* questions.  
(2) Each question carries **eight** marks.  
(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

- 11.** (a) Explain Line & Staff Organization with the aid of neat sketch. State the advantages and disadvantages.

**(OR)**

- (b) Explain the partnership organization and state its advantages and disadvantages.

- 12.** (a) Explain the concept of product development with a block diagram.

**(OR)**

- (b) Explain grounding, shielding and guarding techniques.

- 13.** (a) Explain bath tub curve.

**(OR)**

- (b) Estimate power supply requirements of an electronic product.

- 14.** (a) What is the need for demand survey and market survey? How is it conducted?

**(OR)**

- (b) Explain the concept of Make in India : Zero Defect and Zero Effect.

- 15.** (a) Explain the application of IoT in the context of smart factory and smart manufacturing.

**(OR)**

- (b) Explain the application of IoT in the context of Social Networks.

## PART—C

10×1=10

- Instructions :** (1) Answer the following question.  
(2) The question carries **ten** marks.  
(3) Answer should be comprehensive and the criterion for valuation is the content but not the length of the answer.

- 16.** Analyze the temperature extreme testing for linear and step stress profiles and vibration and temperature cycling with necessary diagrams.

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